

Presentation Plan: Temporal Leakage Poisoning in Concept-Based OPE

ELEC70122 — Monday 16 March 2026

1. Presentation Structure (7 minutes)

Section	Time	Rubric Weight
Introduction & Motivation	1 min	/20
Related Work & Gap	1 min	/20
Methodology	1.5 min	/20
Results	2 min	/20
Conclusions & Limitations	1 min	/20
Buffer/transitions	0.5 min	—

Delivery/visuals (/20), teamwork (/10), and individual Q&A; (/20 each) assessed separately.

2. How to Present Results

Panel Layout (follow CPDIS paper style):

- Left: Primary metric vs horizon (OPE estimate or error)
- Middle: Variance or secondary metric vs horizon
- Right: Diagnostic plot (concept quality/probe R^2)

Slide 1 — Gridworld Results: 3-panel figure, annotate breakdown point with vertical dashed line

Slide 2 — MIMIC Results: Same structure for direct comparison

Slide 3 — Adaptive Gating: Before/after with MixCEM, show gate closing at later timesteps

3. Immediate Action Items

Saturday:

- Finalise Gridworld figures (teammate working on this)
- Draft slide deck skeleton (7-8 slides max)
- Assign speaking parts

Sunday:

- Integrate MIMIC results (partial results OK — say 'preliminary')
- Practice run-through — time strictly
- Prepare Q&A; responses

Monday morning:

- Final practice at 6.5 min (leaves buffer)
- Check tech — slides load, figures render

4. Slide-by-Slide Outline

Slide 1: Title — 'Temporal Leakage Poisoning in Concept-Based Off-Policy Evaluation', names, date

Slide 2: Motivation (1 min)

- OPE critical for safety-critical domains (can't deploy untested policies)
- Sequential OPE suffers from variance explosion (importance weights compound)
- CPDIS reduces variance using concepts — but is it safe?
- Gap: No one examined whether concepts introduce compounding bias

Slide 3: Related Work & Gap (1 min)

- Standard OPE: PDIS, Doubly Robust (high variance at long horizons)
- Concept-based OPE: CPDIS (Majumdar et al.) — reduces variance via abstraction
- Your gap: Soft concepts on stale/forward-filled data leak temporal info → multiplicative bias

Slide 4: Methodology (1.5 min)

- Exp 1: Probe R^2 to detect temporal leakage
- Exp 2: Truncated CPDIS estimates over horizon
- Exp 3: Adaptive gating via MixCEM to suppress poisoned concepts
- Environments: Windy Gridworld (controlled) + MIMIC-III (real)

Slide 5: Results — Gridworld (1 min)

- 3-panel figure; Key finding: Soft concepts degrade after horizon X, hard stable

Slide 6: Results — MIMIC (0.5 min)

- Highlight concept quality collapse in right panel

Slide 7: Results — Adaptive Gating (0.5 min)

- MixCEM gating recovers performance — your novel contribution

Slide 8: Conclusions & Limitations (1 min)

- Contribution 1: First identification of temporal leakage poisoning
- Contribution 2: Adaptive gating as solution
- Limitations: MIMIC Demo small (100 patients), gating needs validation, untested >40 steps

5. Q&A; Preparation

Question	Prepared Answer
How does this relate to distribution shift?	Temporal leakage is concept drift — representation meaning shifts over trajectory (Lecture 3)
Why not just use hard concepts?	Hard may discard useful info; soft more expressive but vulnerable. Gating gives best of both.
Is 100 patients enough?	Sufficient to demonstrate phenomenon exists; full validation needs 40k patients
How define gating threshold?	Entropy-based per MixCEM — gate closes when concept uncertainty exceeds threshold
Connection to conformal prediction?	Both handle distribution shift; conformal uses weighted exchangeability, gating analogous

6. Team Coordination

- Everyone speaks during 7 minutes (even if just one slide)
- Each person knows full methodology for Q&A;
- Designate 'results expert' and 'methodology expert' for Q&A; backup